

A Tale of Five Decoders.

David Ponarovsky

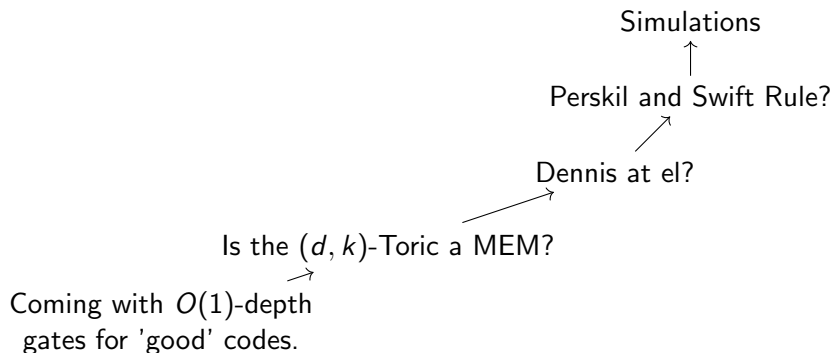
November 3, 2025

Introduction

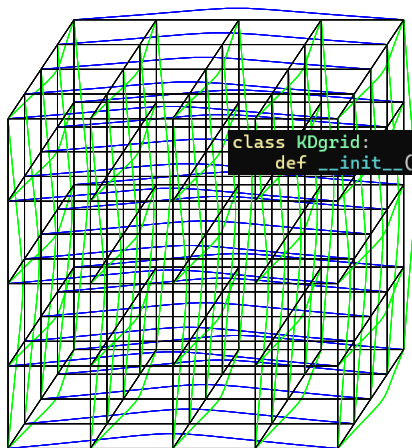
Today:

- ▶ Recap, and where we stood before the summer.
- ▶ Simulations Results.
- ▶ Where we heading.

Map

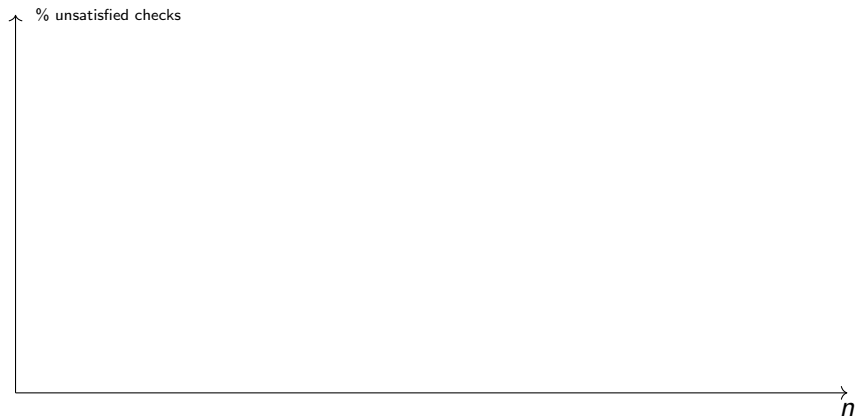


The Code



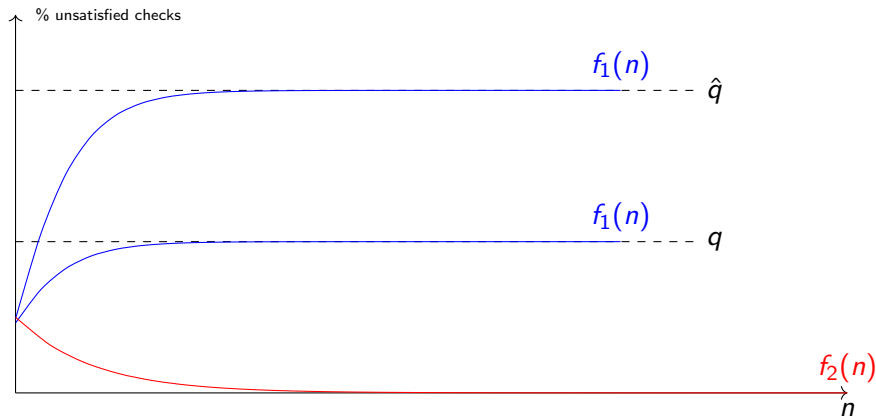
```
class KDgrid:  
    def __init__(self, n, k, celldim=2, checksdim=0,
```

How to Read



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Ideal Result.



colors

In red, No accumulation of errors. In blue decoding in the presence of noise.

Majority

Alg.

Each bit looks at its neighbour checks, if the majority of them is unsatisfied, then it flip itself.

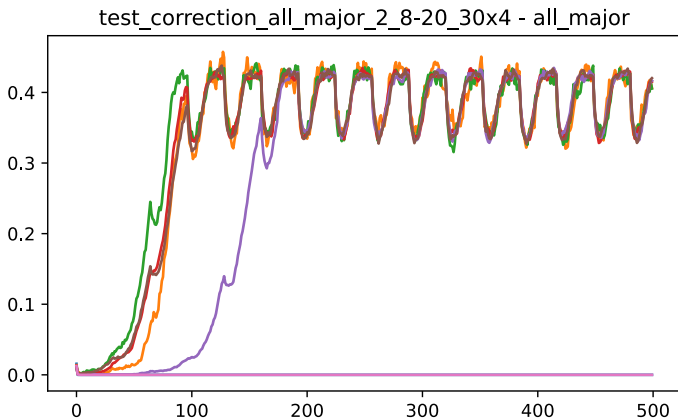
Justification.

Was suggested in Dennis.

Bottom line.

- ▶ Fails to correct, even in the absence of noise.
- ▶ Collisions.
- ▶ Gives better result when requiring $(\frac{1}{2} + \mu)$ -majority.

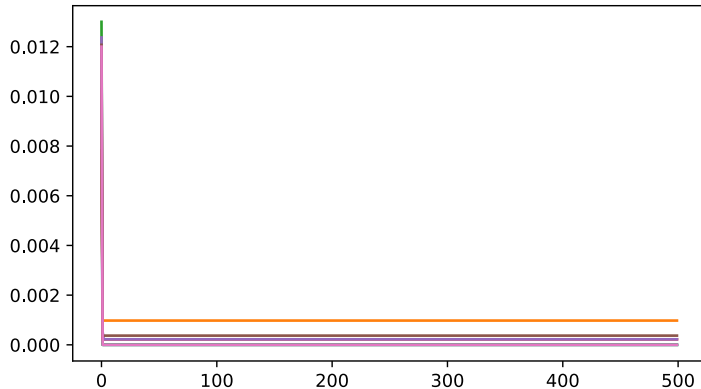
Majority



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Majority

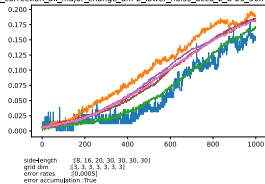
test_correction_all_major_change_diff-2_2_8-20_30x4 - all_major



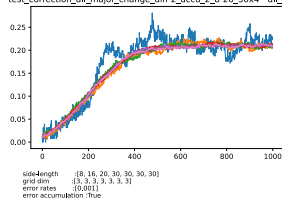
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grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Majority

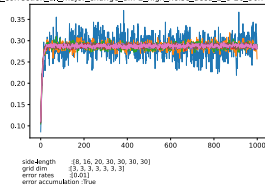
test_correction_all_major_change_diff-2_lower_noise_accu_2_8-20_30x4 - all_major



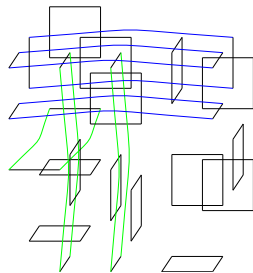
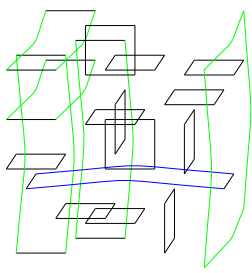
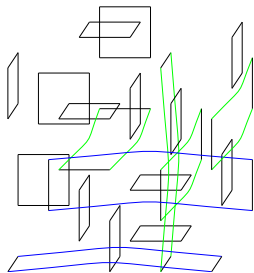
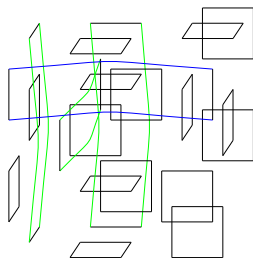
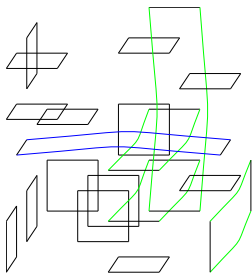
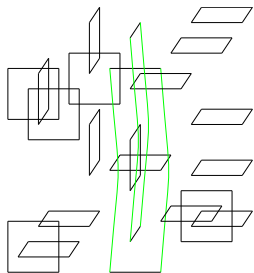
test_correction_all_major_change_diff-2_accu_2_8-20_30x4 - all_major



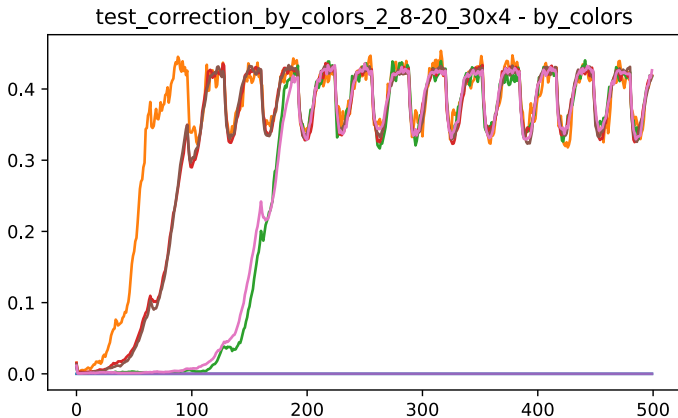
test_correction_all_major_change_diff-2_high_noise_accu_2_8-20_30x4 - all_major



Colors

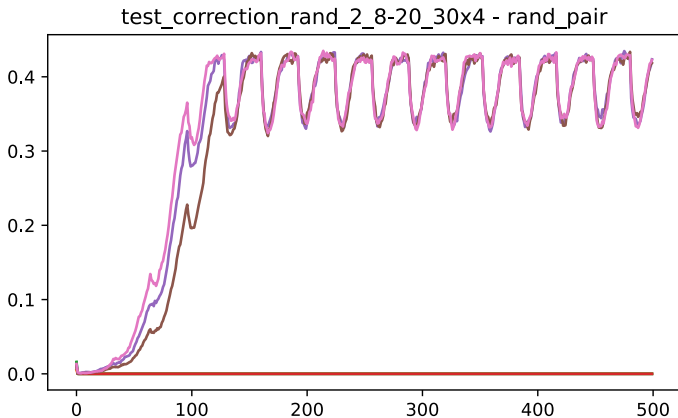


Colors



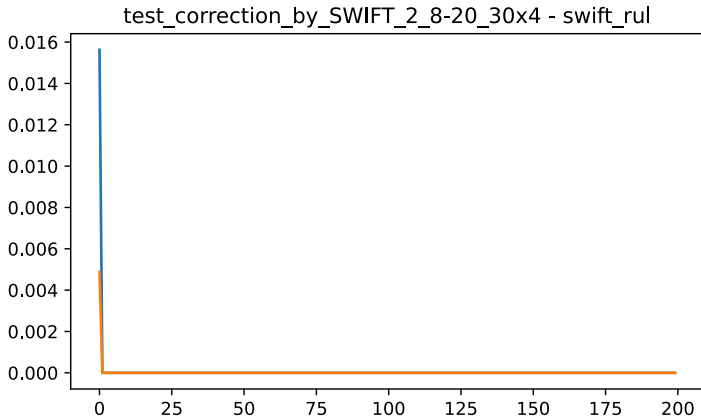
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random



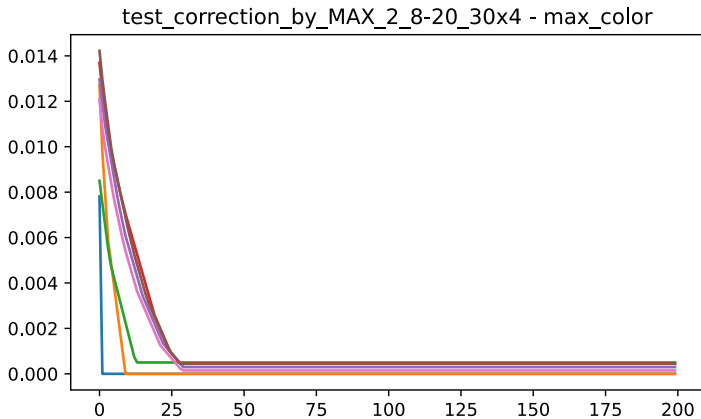
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grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

SWIFT



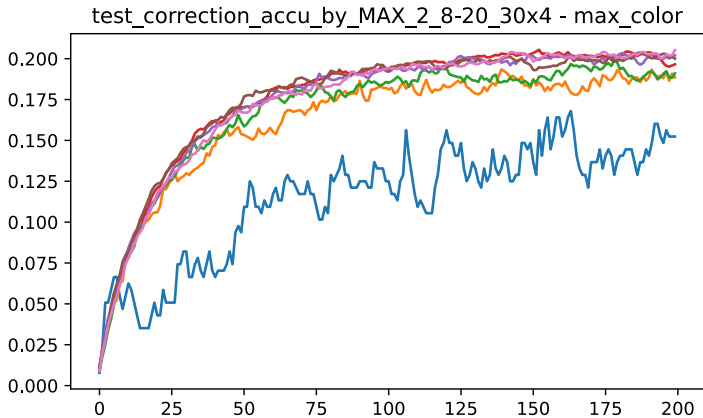
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True